

REVIEW

Bioseparations and Downstream Processing

Managing integrated continuous bioprocesses in real time: Deviations in product quality

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Abstract

The N-mAb case study was produced by the National Institute for Innovation in Manufacturing Biopharmaceuticals (NIIMBL) to support teaching and learning for both industry and regulators around adoption of advanced manufacturing process technologies such as integrated continuous bioprocesses (ICB) for monoclonal antibodies (mAbs). N-mAb presents the evolution of an integrated control strategy, from early clinical through process validation and commercial manufacturing with a focus on elements that are unique to ICB. The entire N-mAb case study is quite comprehensive, therefore this publication presents a summary of the chapter on managing deviations from a state of control in real time. This topic is of critical importance to ICB and is also applicable to batch processes operated at a rapid cadence.

1 | INTRODUCTION AND SCOPE

The purpose of this publication is to discuss considerations for managing deviations to critical controls in the context of integrated continuous bioprocesses (ICB).¹ Critical controls are defined here to include critical process parameters (CPPs), critical material attributes (CMAs), critical in-process controls (IPCs), and drug substance specifications. Although, under cGMP, critical controls also apply to facilities, instruments, personnel, utilities, etc., the general cGMP considerations for deviations that impact these controls are not unique to ICB and are not part of this publication.

In general, deviations should be opened whenever there is a departure from a state of control, and this article will focus on a subset of potential triggers for deviations:

- Excursions of CPPs from registered operating ranges with the potential to impact critical quality attributes (CQAs)
- Excursions of critical IPCs from registered action or reject limits

This publication covers key technical and Quality Management System (QMS) considerations and requirements that ensure

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investigations and associated actions are effective for ICB. QMS requirements include the following:

- Identification, segregation and/or disposal of potentially impacted drug substance lots or constituent portions of these lots
- Rapid root cause analyses (RCA) and product impact assessments (PIA) for which the time available to investigate and the access to in-process samples for non-routine investigational testing will be constrained by the ICB
- Immediate corrective actions required to continue processing additional portions of the lot

The article will not cover details of lot disposition decisions or post-manufacturing corrective and preventive actions (CAPAs) because their decision framework and impact to operations are not unique to ICB.

Critical material attributes (CMAs) are an important consideration for ICB because of the intensified use of critical single-use systems. Deviations to CMAs would normally be detected off the critical path for ICB decisions, either well in advance of use, or after the fact in the case of undetected defects. Therefore, CMAs will not be discussed in depth in this publication; for additional details refer to the N-mAb case study. Finally, the CPPs and

CQAs related to adventitious agent control are also excluded from this publication, but considerations can be found in Chapter 5 of N-mAb.

1.1 | Framework considerations for the N-mAb process

This discussion of deviation management for the N-mAb process is dependent on the N-mAb process flow framework as described in Figure 1. Note that the number of in-process pools available for representative sampling of the process as well as the potential for holding the process is significantly reduced for ICB. Several aspects of the ICB framework are unique or intensified relative to batch processing, and these aspects may require QMS adjustments to address their challenges. The ICB framework poses constraints of *limited time* and *limited access*, and it enhances the potential for *comingling* of impacted portions of a lot with unimpacted portions of a lot which is inherent to continuous processing to a certain degree due to variations in the overall residence time distribution across the process.^{2,3} (See Figure 2). At a minimum a deviation management system must inform the disposition of the resulting drug substance batch. The management of product quality deviations requires at least three elements:

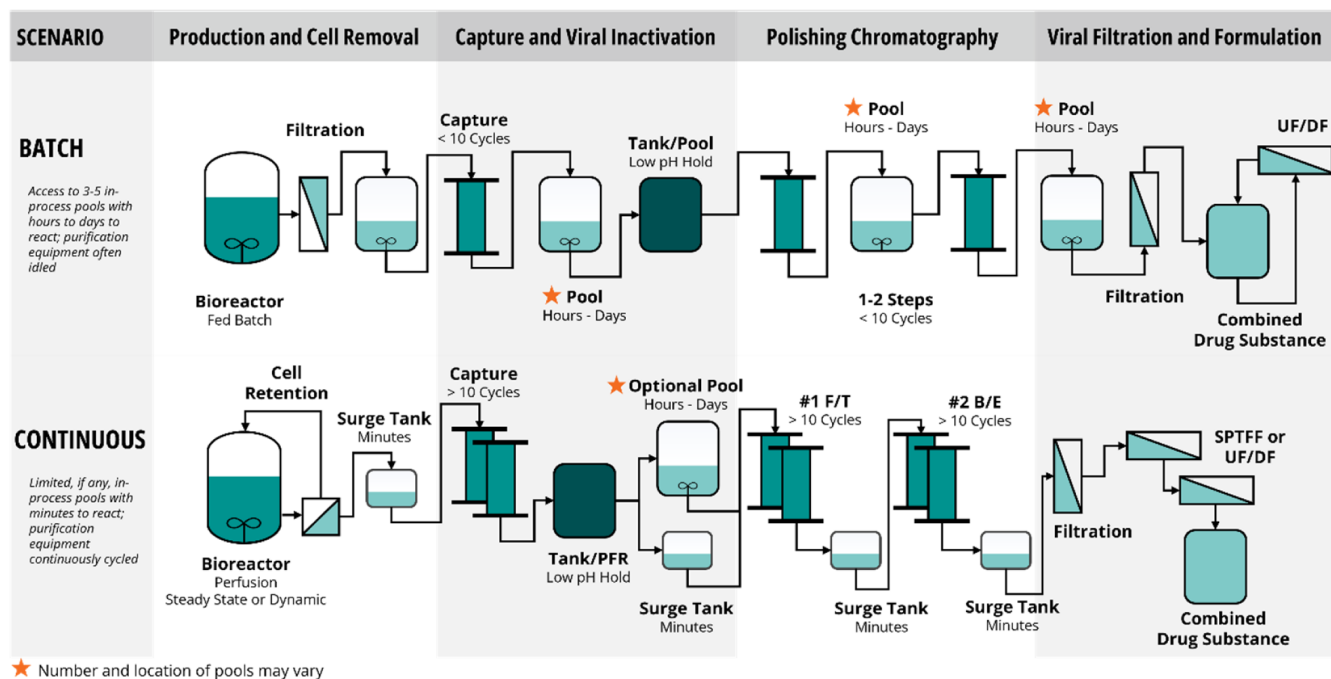


FIGURE 1 Framework comparison of batch and continuous biologic manufacturing. Batch and continuous processing frameworks differ principally in the presence or absence of intermediate pool hold steps, in the number of processing cycles, and in operating modes of certain steps (e.g., bioreactor, cell removal, etc.). Continuous processing could deploy a lean or hybrid approach. In the lean approach processing is truly continuous with no intermediate pools and thus IPC testing of CQAs is of limited value unless performed using in-line, rapid PAT technologies. Absent PAT, in-process sampling and off-line testing can be performed retrospectively and can provide supportive data for deviation investigations. The hybrid approach involves one or more intermediate pools or intervention options (one example highlighted above), and thus IPC testing can be used to intervene and remove impacted process flow or pools from the final drug substance. This approach is dependent upon the timely availability of the IPC result to enable a response to a deviation in real time and possibly in-process stability data to support longer hold times if a longer response time is required.



FIGURE 2 Factors impacting deviation management in integrated continuous bioprocesses. The following factors intensify the challenges of managing deviations in ICB. *Limited time* to investigate root cause or intervene to prevent further impact. Rapid, risk-based decisions may be required to either forward process, divert to waste, or discontinue processing. *Limited access to samples for forensic testing.* The potential exists to include slip-stream sample collection, but this would introduce complications in sample management, testing strategy, and data interpretation, and is unlikely to be incorporated. *Comingling* of an impacted portion with the unimpacted remainder may be inherent to continuous processing. It is not acceptable to deliberately blend non-conforming material to meet specifications, but comingling may be permissible, subject to a PIA within the framework of the QMS.

1. Detection of an excursion beyond a defined limit that potentially impacts product quality for either a critical process parameter (input), a critical process control (output), or an acceptance criterion for drug substance quality
2. An investigation to determine the root cause(s) for the loss of control
3. A determination of the potential impact of the excursion on the resulting drug substance batch and any other impacted batches associated with the root cause(s)

In addition, for ICB, a fourth element is expected to be an integral part of the deviation management system, which is dependent on the specific process framework design.

4. An intervention to divert a portion of the process stream (or batch) to waste that is predetermined to likely be negatively impacted by the detected deviation, to preserve the remainder of the batch

An intervention point is a location in the process stream where a deviation detected by the outcome from process monitoring or sample testing can result in an action to the process flow. The minimal actions available at an intervention point are to proceed with further processing at risk or divert to waste the appropriate amount of the process stream that has the potential to be impacted by the deviation.

With considerations of the N-mAb framework process design, diverting to waste can be active (prospective) or static (retrospective). The intervention to divert to waste a portion of an active process stream (prospective diversion) is considered the basic approach to be taken with ICB. However, if the framework incorporates intermediate batch pools, alternate approaches that include segregation of material, confirmation

testing, and re-introduction for forward processing (retrospective diversion) can be considered if the systems and QMS enable these methodologies. Additional discussion of this approach can be found in the original N-mAb publication.

The active management of a CPP deviation will be determined based on the integration of monitoring instruments and equipment, an understanding of the response time for detection with respect to potential action, and the capability to remove the impacted process stream from the system at the relevant intervention points. A risk-based assessment of the potential impact to the final batch is required to decide how to handle the portion of the batch impacted by the CPP deviation. Possible outcomes include diverting to waste, forward processing at risk or segregating material for potential re-processing or re-entry into the process. Table 1 presents an example deviation management strategy for addressing CPP deviations. This table lists the potential CPPs as well as considerations for defining and detecting a deviation and the immediate/potential actions to be taken.

1.2 | Framework and control strategy implications related to excursions to in-process controls (IPC)

Sampling of in-process material for CQA testing is used for process monitoring of critical unit operations controlling the CQA. For example, if two orthogonal steps are required to control a CQA, then an IPC for monitoring the first step may be desirable to provide more granular monitoring, control, and deviation management.

For this discussion, each IPC is expected to have defined limits along with established actions to be taken if those limits are exceeded. In addition, IPC trending should be used to identify process performance drift and the QMS should pre-define actions to investigate any drift appropriately. The specific limits and actions should be informed from process characterization studies that established the process operating targets and limits.

For ICB, the use of IPCs to detect and react to a loss of control is dependent on the processing framework and associated control strategy. In addition, the use of PAT in ICB as an element of the overall control strategy is dependent on the availability of robust and readily available technologies that can provide outputs of CQAs in a time frame that allows active management of the process. Two approaches can be taken into consideration when establishing an appropriate control strategy (See Figure 1):

1. A *lean approach*, where the framework design is truly continuous with no intermediate pools and thus IPC testing of CQAs is of limited value unless performed using in-line, rapid PAT technologies. In this instance, CQA testing is performed on the final drug substance only.
2. A *hybrid approach*, where the framework design involves intermediate pools, or intervention options, and thus IPC testing can be used to intervene and remove impacted process flow or pools from the final drug substance.

Other considerations for the use of IPCs to manage deviations may depend on whether IPC testing is considered part of a dynamic

TABLE 1 Examples of process parameter controls and deviation actions.

Step	Process parameter	PAR	NOR	Monitoring strategy (Discrete or continuous)	OOR duration triggering a deviation	Action to be taken immediately (Pause, Continue, or Divert)	Data and/or samples available for investigation (Step output, subsequent step output, DS testing)
AEX	Load pH	6.8–7.2	6.9–7.1	Continuous (post-VI neutralization)	Outside of NOR (inside PAR) for ≥ 0.2 CV or outside of established range for 0.05 CV	Continue processing. Impacts DNA levels and there is redundant clearance from subsequent steps	VI inactivation pH (root cause investigation) VI protein concentration (root cause) AEX (DNA non-routine) CEX (DNA non-routine) DS is not tested for DNA
CEX	Load pH	6.8–7.2	6.9–7.1	Continuous (post-load adjustment)	Outside of NOR (inside PAR) for ≥ 0.2 CV or outside of established range for 0.05 CV	Divert (prior to loading) Impacts DS aggregates levels and this is last step to clear	CEX product (aggregates non-routine test) DS (aggregates routine test)

operational feedback approach or a requirement prior to further forward processing, especially if intermediate batch pools are part of the framework design.

If the framework option omits any intermediate pools that represent the entire drug substance batch, then the control strategy would likely rely predominantly on monitoring and control of CPPs in lieu of measuring IPCs. In cases where process characterization studies have established the linkage of IPC sampling frequency and location to final drug substance attributes, then IPC testing using point-in-time sampling can be incorporated. However, management of in-process testing for CQAs would require automated at-line sampling and analytical instrumentation and would add complexity to the control strategy.

If the framework option selected includes a homogenous, intermediate pool representative of the entire drug substance batch, it would be feasible to sample and test this pool for conformance of a CQA to an IPC limit. In such a situation, the implications of a deviation from the IPC limit would default to conventional batch processing scenarios. As the entire batch would be contained in the intermediate pool, it is unlikely that there would be any incentive to divert the pool to waste to prevent product impact to the batch. Instead, unless it is grossly contaminated or otherwise defective, the material can be forward processed through the remaining unit operations, and the investigation and lot disposition activities could proceed decoupled from ongoing operations.

It is expected that deviations associated with a potential safety risk (e.g., pH out of operating range during viral inactivation for greater than a pre-defined time frame) will be predetermined to result in immediate diversion to waste from the process flow to minimize any impact to the DS batch.

1.3 | Deviation investigation requirements

A deviation from established controls must be investigated prior to disposition of the lot. The investigation may assess root cause, product impact, and the need for any CAPAs.

While the root cause and/or product impact may be immediately apparent based on routinely collected information, in some cases, additional data collection may be required to reach a conclusion.

Confirmed excursions outside of IPC reject limits or confirmed OOS events typically result in rejection of the entire lot or of the impacted portion of the lot. In contrast, excursions outside of IPC control or action limits or outside of CPP acceptable ranges may require a RCA and PIA, but they do not necessarily result in lot rejection.

1.4 | Adjustments to the quality system required with ICB

As mentioned earlier, and described in Figure 2, the constraints of limited time and limited access combined with the potential for comingling represent a significant challenge for deviation management. To manage these factors the QMS may be enhanced by reference to documented and approved heuristic algorithms to facilitate rapid decisions. These algorithms may address three orthogonal characteristics following a deviation to the defined process controls:

1. How does the severity and duration of the CPP or IPC excursion relate to the proven acceptable range(s) (PARs) of the parameter(s) or action/reject limits of the IPCs?
2. Is the root cause clearly assignable?
3. What is the proportion of the lot impacted by the loss of control?

The severity and duration of the CPP or IPC excursion informs the likelihood of product impact; the root cause identification informs confidence in the ability to restore control; and the proportion of lot impacted informs whether the quality of the lot is acceptable. As described in Figure 3, the QMS algorithms should consider these factors to drive the investigation and any decisions to intervene in processing operations. For example, detection of a deviation could trigger an increase in sampling and testing frequency to limit the amount of material diverted to waste in the case of a transient or short-term excursion followed by a return to control conditions.



EXCURSION SEVERITY & DURATION

Informs potential for product impact



ROOT CAUSE IDENTIFICATION

Informs ability to restore control



PROPORTION IMPACTED

Informs overall impact to comingled lot

FIGURE 3 Factors informing quality management system decision algorithms in integrated continuous bioprocesses. The QMS should consider the following factors to inform investigation elements and decisions to intervene in processing operations. *Severity and duration* of CPP or IPC excursions inform likelihood of product impact and a decision to divert to waste. Temporary CPP excursions outside of nominal PARs may be considered acceptable within characterized bounds defined by extent and duration. For IPC excursions inside of reject limits, or CPP excursions within the PAR but outside of expected control ranges, a deviation should be triggered, but it should not be necessary to divert product to waste. *Root cause identification*. If the root cause is related to an obvious malfunction or error and can be quickly corrected the decision to continue processing can be supported. The root cause should also inform the potential for product impact. If the root cause is not immediately apparent, there will be less confidence about restoration of control or potential for no product impact. *Proportion of the lot impacted*. Comingling of the impacted portion may occur inherent to continuous processing. In the event of a transient excursion impacting only a few processing cycles, the quality of the drug substance may potentially be considered acceptable, depending on the proportion impacted and subject to QMS considerations. A more persistent loss of control could result in significant impact to the overall quality of the lot.

1.5 | Supportive data and documentation

The aforementioned considerations may be integrated for decision support by access to a database of relevant, up-to-date experimental and process data, which is the foundation for establishing PARs. This database should remain a living entity, updated with new learnings from investigations, additional studies, etc. Note that some companies use certain electronic tools for consolidating data that supports criticality determinations or data management for providing structured content for regulatory submissions⁴; these tools could be adapted to support extended applications of a process information database and could also serve as pre-requisites for automated decision support, discussed in *Discussion and Future Directions*.

1.6 | Scenarios for managing deviations

The “immediate” nature of ICB imposes significant time constraints on the management of deviations and the resulting decisions. The ability to take rapid, but relevant, triage decisions regarding forward

processing, diversion to waste, or holding of an intermediate will strongly depend on the quality of the process characterization package. Indeed, the availability of robust CCP-CQA linkage models will support the implementation of rapid, heuristically driven RCA and PIA, and the definition of pre-established decision algorithms minimizing unnecessary disruptions to manufacturing operations:

Root cause considerations: The scope of the RCA, and the resulting processing triage decision, will depend on two main considerations:

- Is the root cause of the excursion immediately assignable?
- Is this root cause fully reversible prior to the next processing cycle?

When the root cause of an excursion is immediately assignable, and this root cause is determined to be reversible prior to the next processing cycle, then a decision to continue processing could be considered. In the case of excursions not immediately reversible, the material produced may have to be segregated until CAPAs are implemented.

When the excursion has no immediately assignable root cause and no obvious reversibility, run termination may be necessary and a deliberate RCA should be conducted. Even when a proximate root cause can be identified and corrected, it may be necessary to conduct a formal RCA to confirm the ultimate root cause (e.g., training, materials, procedures, etc.) and determine what appropriate CAPAs should be implemented prior to resuming manufacturing.

Product impact considerations: The QMS should address the product impact implications of an excursion and the triage decisions taken following RCA. A formal PIA may not be required for a portion of a lot affected by an excursion when it is diverted to waste. Nevertheless, an assessment may still be required for other portions of the lot potentially associated to the findings of the RCA. The PIA should consider the following:

- How severe is the excursion (amplitude, duration)?
- Where in the process did the excursion occur?
- Can downstream steps compensate for the excursion?

In the case where the lot portion is forward processed, the quantitative assessment of the impact of the excursion on product quality may depend on multiple factors. Hence, the true product impact may remain unknown until confirmatory in-process (routine or non-routine) or release testing results are obtained.

In summary, for ICB, it is essential that an agile Quality mechanism and robust data package exist to drive timely decisions about the scope of the RCA and the evaluation of its impact to ongoing production. The considerations and decisions involved in managing deviations in ICB are shown in Figure 4. To illustrate this process, three scenarios were considered for managing a deviation.

1.6.1 | Scenario 1: Deviation detected via monitoring of CPPs

Process control strategies for ICB will heavily rely on CPP monitoring. Aside from CPPs tied to adventitious agent controls (e.g., viral

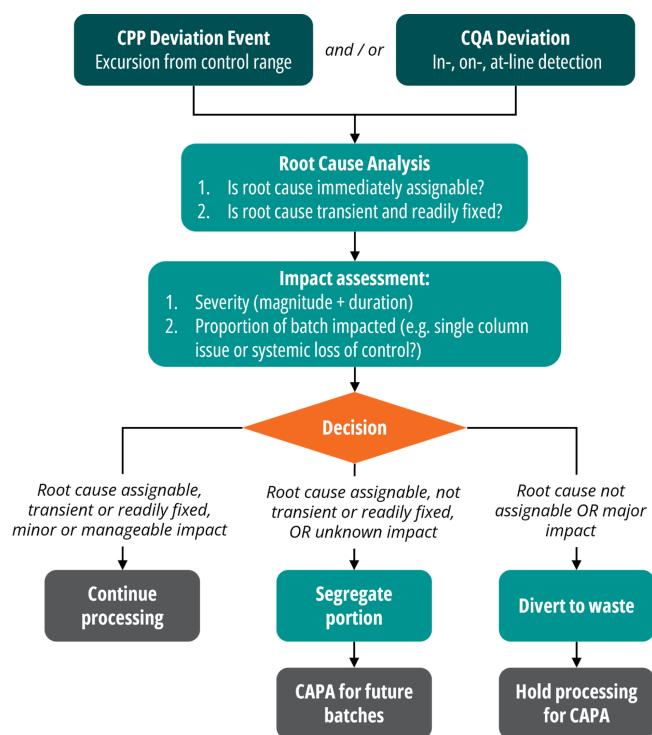


FIGURE 4 Considerations for critical process parameter or critical quality attribute deviations in an integrated continuous bioprocess framework.

clearance), most CPP excursions do not constitute automatic causes for lot rejection. As discussed previously, the severity and recurrence of the excursion and proportion of the lot impacted as PIA is performed will support decision making on whether or not to terminate a lot, and may inform the investigation scope for RCA following the QMS guidelines.

The availability of extensive CCP-CQA linkage models is again required to support rapid decision making, particularly in cases where the quantitative impact of the CPP excursion on a CQA depends on multiple factors (e.g., interactions with other CPPs). Often, the product impact of a CPP excursion may remain unknown until confirmatory CQA testing results are obtained, and the affected lot portion may therefore have to be segregated. If the CQA impacted by the CPP excursion is associated to a drug substance release specification, then a passing specification result may inform the lot disposition decision, subject to any additional product impact and QMS considerations.

Table 2 lists a few examples illustrating the integrated QMS decision framework considering severity, root cause, and proportionality of CPP deviations.

1.6.2 | Scenario 2: Off-line detection of deviation to CQAs

Off-line analytical testing of CQAs typically occurs in a QC laboratory, and results may be reported from hours to weeks after sample

collection. In the ICB framework, the portion of the lot impacted by a CQA excursion will likely have been forward processed to a drug substance batch before results become available.

When a CQA deviation is detected in a “whole lot” pool IPC or drug substance release testing, the PIA may proceed according to conventional QMS practices applied to batch processing. The entire batch will either be accepted or rejected based on the findings of the investigation. Note that there may be no representative “whole lot” pools in some versions of the N-mAb framework, and hence CQAs would only be assessed through final testing of drug substance.

Similarly, the RCA will proceed long after the initial deviation occurred and cannot have any immediate implications for the production of that batch. The Quality and Manufacturing Units will need to reach a decision to either continue operations at risk or terminate the production run.

1.6.3 | Scenario 3: Real-time detection of deviations to CQAs

Emerging technologies permitting rapid on-, in- or at-line testing (collectively referred to as “real-time” detection) facilitate continuous or intermittent monitoring of CQAs. Real-time results can inform the decisions from observed deviations and could also be used to feed-forward or feed-back for adaptive control strategies, hence reducing the potential risks associated with forward processing.

When the characterization package enables a clear cause and effect linkage between the real-time detection of a CQA deviation and a concomitantly observed CPP control excursion, a rapid PIA and material segregation decision may be taken. In the absence of a correlated CPP excursion, or when a correlated observed CPP excursion has only a plausible linkage to the CQA, the RCA must proceed with very limited insights into the CQA excursion. The triage decision may be further informed by whether the action limit excursion is transient or sustained. However, if the CQA exceeds a rejection limit with no known root cause, it is advisable to divert to waste and pause operations at the point of control for the CQA to permit a comprehensive investigation and appropriate CAPAs.

Consequently, real-time CQA monitoring capability would enable more agile decision-making by the QMS in an ICB environment, including considerations for whether deliberate co-mixture in downstream processing is acceptable. Examples illustrating the integrated QMS decision framework considering severity, root cause, and proportionality of real-time CQA deviations are described in Table 3.

2 | CONCLUSIONS AND FUTURE DIRECTIONS

The N-mAb case study has attempted to present an overall story of the development of an ICB from process design through commercial manufacturing to support teaching and learning for both industry and regulators around adoption of advanced manufacturing process

TABLE 2 Scenarios for managing deviations to critical processing parameters.

Critical control point	CPP excursion	CQA Potentially Impacted	Detectability of potential CQA impact	Severity of impact (extent, duration, proportionality)	Root cause assignable?	Deviation management considerations (including confirmatory QC tests)
Bioreactor	Bioreactor pH	Glycosylation, HCP, Aggregates	Glycosylation: No routine test HCP: DS release test Aggregates: DS release testing	Transient excursion outside NOR, inside PAR Prolonged excursion outside PAR	Yes No	Immediate: Forward process at risk Long term: Assess need to investigate ultimate root cause Immediate: Divert to waste Medium term: decide if PC data suggest process can recover to SS. If so, resume processing at risk after restoring SS. Long term: Assess need to investigate ultimate root cause
Protein a capture chromatography	Capture column load ratio	HMWS	DS release testing	One cycle exceeds PAR	Yes	Immediate: divert to waste or forward process to polishing depending on severity and proportionality
Polishing chromatography	Polishing elution buffer ionic strength	HMWS	DS release testing	One cycle outside NOR, inside PAR	Yes	Immediate: forward process at risk Long term: Assess need to investigate ultimate root cause

technologies for mAbs. The current publication summarizes elements of an effective ICB QMS to detect deviations in process parameters and/or product quality in real-time, investigate the root cause, evaluate the cause/effect relationship, and make a risk-based and data-driven decision to forward process or divert to waste a suspect bioprocess stream segment. IBC bioprocess designs with and without in-process intermediate pools were discussed that highlight different CPP, IPC, and final batch testing strategies for managing the decisions associated with deviations. Both IBC scenarios require segment disposition decisions to be made with limited access to representative in-process materials on a timescale that prevents unwanted co-mingling of potentially at-risk material. Indeed, a truly effective PAT strategy requires the seamless interplay of reliable in-line sensors and instruments to enable real-time measurements and establishment of a robust digital environment to securely manage the flux of data generated. Deviation management decisions ultimately rely on a quantitative understanding of the relevant impact on CQAs. The QMS may therefore benefit from continuously updated process understanding and associated heuristic algorithms that can consider the severity (attribute criticality, extent/duration, batch proportionality) and assignability of the root cause to inform a risk-based deviation management decision.

The QMS associated with ICB is expected to evolve to accommodate multi-variate design space for detection and management of deviations to allow more flexible operations within a non-orthogonal combination of CPP acceptable ranges.^{5,6} One can also envision sophisticated ways of linking data supporting criticality assessments such that it can be readily and quickly retrieved.⁷⁻⁹ As a result, ongoing learnings from investigations, additional experimentation, etc. may be included as part of the evolution of the design space. These design space updates could simultaneously consider magnitude and duration of an excursion. Operationalizing such design spaces may require the alarm systems, batch record, and QMS to access computational tools to determine if a deviation from the multi-variate design space has occurred, thus opening opportunities for artificial intelligence applications that enable automated decision support and more streamlined decisions when deviations are detected (See Figure 5).

Implementation of artificial intelligence or machine learning (AI/ML) to rapidly evaluate the data collected using multivariate models and make *automated recommendations* to either divert or forward process becomes an appetizing “first step” toward fully automated process control. Automated decision support tools could integrate deviation signals from non-conforming CPPs, supportive information about conforming CPPs, design space knowledge, and decision heuristics to provide recommended immediate steps. Such tools should be validated to provide reliable outputs for pre-defined scenarios, and these outputs could be referenced immediately on the manufacturing floor to drive real-time decisions. A second evolution would be to relinquish control of specific process skids where successful AI/ML recommendations have been well documented, and *automated decisions* to divert/forward process could be carried out within the limitations of the overall process dynamics (unit operation residence time, skid reaction delay, etc.). When the automated support tool is unable to map a scenario to the predefined knowledge

TABLE 3 Scenarios for deviation management with real-time critical quality attribute monitoring.

CQA Impacted	Point of detection	CQA Impact (extent, duration, proportionality)	CPP or material attribute causal factor assignable?	CPP impact (if identified) (extent, duration, proportionality)	Deviation management considerations (including confirmatory QC tests)
Aggregation	Post CEX B/E Polishing Surge Vessel	Reject limit, recurrent for 1 column cycle	No	N/A	<i>Immediate:</i> Divert to waste <i>Medium term:</i> Investigate root cause; if one column identified, take off-line and operate with reduced yield.
			Yes (Resin Material Attributes)	Yes—reduce elution pH PAR to achieve robust performance	<i>Immediate:</i> Divert to waste <i>Medium term:</i> Investigate root cause; if one column identified, take off-line and operate with reduced yield. RCA identifies variation in lot-to-lot resin material attributes that could result in tighter acceptance criteria for resin lots or, if that is not possible, modification of the CEX elution conditions could result in robust performance over the variation expected from multiple resin lots.

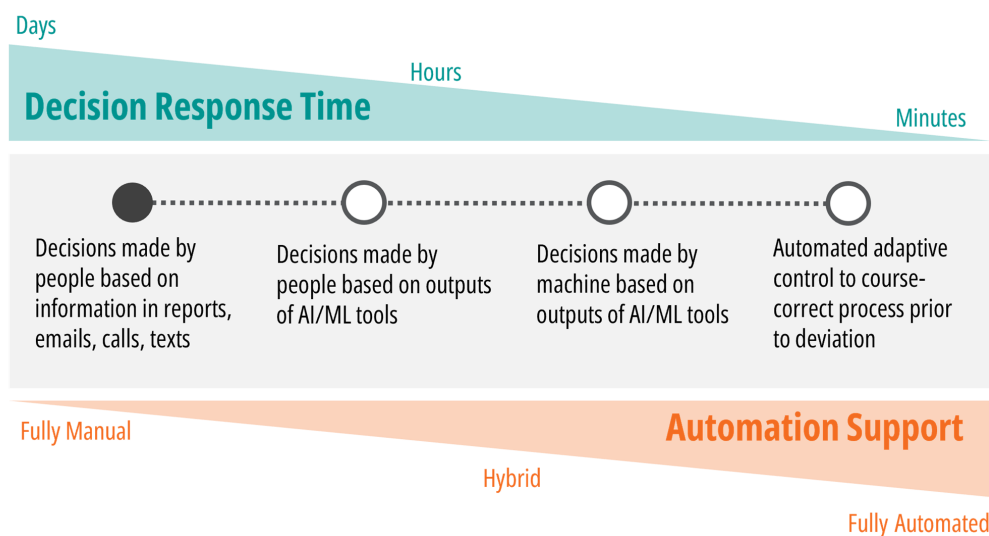


FIGURE 5 Evolution of decision support added by digital transformation. The current state for most organizations is at the far left in this diagram: decisions regarding excursions are made by people using information gathered by traditional means. As the cadence of processing increases, for example, for an integrated continuous bioprocess, the pace of the process will require faster decisions. Faster decisions can be enabled by increased integration of real time data management aided by AI/ML tools to either support decisions made by people or, in the more advanced cases, drive automated responses.

space, the issue would likely be escalated for a more deliberate, expert-driven assessment.

We anticipate an increased role for automation, including AI/ML, in QMS systems as technology advances. Advanced control strategies described herein may still be too forward-looking for systematic implementation at this time and could require advancing the analytical technologies beyond what is currently available. It is important, however, to keep these end goals in sight as instrument and control capabilities continue to advance. Over time, extensive data on process steps where product CQAs are created or modified and the

process conditions constituting critical control ranges for each CQA will be generated. Concurrently, continuously evolving analytics can enable a more in-depth understanding of how the CPP/CMA variations affect the product CQAs for each unit operation. Given the ability to collect and organize that information over the product lifetime, it seems feasible that AI/ML-driven digitally enabled process understanding can support process control decisions, if not automatically predict and adapt, based on accumulated prior knowledge. We believe that AI/ML will initially be involved in deviation detection and advanced warning, beyond existing process monitoring, and through

comprehensive monitoring of multiple factors, including bioburden, weather, utilities, training logs, increase in numbers of new operators, and personnel movement within the site, in addition to all activities that may elevate the risk of deviations. With time, this involvement may evolve to enable automatic process control through artificial intelligence (AI) without the reliance on operators to make decisions. Eventually, automation may evolve over the next decade to allow for adaptive control or even perform batch release without relying on human operators as computing power increases and leads to better scenario analysis, risk mitigation, and decision making. Regulatory agencies are already considering the impact of AI on the future of biopharmaceutical manufacturing and are assessing the level of regulation and notification expected from applicants to define the impact on product quality.¹⁰

GLOSSARY

AEX	anion exchange
B/E	bind and elute
CAPA	corrective and preventative actions
CEX	cation exchange
CPP	critical Process Parameter
CQA	critical quality attribute
CV	column volumes
DS	drug substance
F/T	flowthrough
HCP	host cell proteins
HMWS	high molecular weight species
ICB	integrated continuous bioprocess
NOR	normal operating range
OOR	out of range
PAR	proven acceptable range
PC	process characterization
PIA	product impact assessment
QC	quality control
RCA	root cause analysis
SS	steady state
VI	virus inactivation

AUTHOR CONTRIBUTIONS

Gustavo Grampp: Conceptualization; writing – original draft; writing – review and editing. **Allen Bosley:** Writing – original draft; writing – review and editing. **Maen Qadan:** Conceptualization; writing – original draft; writing – review and editing. **John Schiel:** Writing – original draft; writing – review and editing. **Andy Spasoff:** Writing – original draft; writing – review and editing. **Pascal Valax:** Writing – original draft; writing – review and editing. **Gene Schaefer:** Conceptualization; supervision; writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

PEER REVIEW

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DATA AVAILABILITY STATEMENT

Data sharing not applicable - no new data generated.

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REFERENCES

1. NIIMBL (National Institute for Innovation in Manufacturing Biopharmaceuticals). N-mAb: A Case Study to Support Development and Adoption of Integrated Continuous Bioprocesses for Monoclonal Antibodies. Version 1. 2022 www.n-mab.org (last accessed May 23, 2023)
2. Lali N, Jungbauer A, Satzer P. Traceability of products and guide for batch definition in integrated continuous biomanufacturing. *J Chem Technol Biotechnol*. 2022;97:2386–2392. doi:[10.1002/jctb.6953](https://doi.org/10.1002/jctb.6953)
3. Senčar J, Hammerschmidt N, Jungbauer A. Modeling the residence time distribution of integrated continuous bioprocesses. *Biotechnol J*. 2020;15:2000008. doi:[10.1002/biot.202000008](https://doi.org/10.1002/biot.202000008)
4. Ahluwalia K, Abernathy MJ, Beierle J, et al. The future of CMC regulatory submissions: streamlining activities using structured content and data management. *J Pharm Sci*. 2021;111:1232–1244. doi:[10.1016/j.xphs.2021.09.046](https://doi.org/10.1016/j.xphs.2021.09.046)
5. Kepert JF, Cromwell M, Engler N, et al. Establishing a control system using QbD principles. *Biologicals*. 2016;44:319–331. doi:[10.1016/j.biologicals.2016.06.003](https://doi.org/10.1016/j.biologicals.2016.06.003)
6. Rathore AS, Singh SK, Kumar J, Kapoor G. Chapter 48 - implementation of QbD for manufacturing of biologics—has it met the expectations? In: Jagschies G et al., eds. *Biopharmaceutical Processing*. Elsevier Ltd; 2018:1051–1073. doi:[10.1016/B978-0-08-100623-8.00048-7](https://doi.org/10.1016/B978-0-08-100623-8.00048-7)
7. Joglekar G, Reklaitis GV, Giridhar A, Mockus L. Knowledge Management in Support of QbD, Chapter 14. In: Reklaitis GV, Seymour CS, García-Munoz S, eds. *Comprehensive Quality by Design for Pharmaceutical Product Development and Manufacture*. American Institute of Chemical Engineers, Inc; 2017:361–386. doi:[10.1002/9781119356189.ch14](https://doi.org/10.1002/9781119356189.ch14)
8. Zobel-Roos S, Schmidt A, Mestmäcker F, et al. Accelerating biologics manufacturing by modeling or: is approval under the QbD and PAT approaches demanded by authorities acceptable without a digital-twin? *Processes*. 2019;7:94 <https://www.mdpi.com/2227-9717/7/2/94>
9. Schmidt A, Frey J, Hillen D, et al. A framework for automated quality assurance and documentation for pharma 4.0. In: Habli I, Sujan M, Bitsch F, eds. *Computer Safety, Reliability, and Security. SAFECOMP 2021. Lecture Notes in Computer Science*. Springer International Publishing; 2021:12852. doi:[10.1007/978-3-030-83903-1_15](https://doi.org/10.1007/978-3-030-83903-1_15)
10. FDA. *Artificial Intelligence in Drug Manufacturing*. 2023 Available from: <https://www.regulations.gov/docket/FDA-2023-N-0487>. (last accessed May 23, 2023)

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